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Transformation

Wellness, part 1: The rise of the wellness economy

27 May 2026

Key takeaways

- While technology drives progress, it also introduces health risks, as constant connectivity and AI-driven tools increase reliance on screens at the expense of in-person interaction. Prolonged screen exposure is linked to poor posture, eye strain, obesity and loneliness, with tech-related health conditions estimated to cost the global economy at least $7 trillion annually.
- Physical and mental health are interconnected, with stress in one area often spilling into the other. As tech-related health risks grow, BofA Global Research emphasizes the rising importance of multidimensional wellness solutions, spanning physical, emotional and social wellbeing.
- According to Global Wellness Institute, the US wellness economy is valued at around $2 trillion - the largest in the world - and accounts for roughly 30% of the global market. With a population of about 340 million, this equates to approximately $6,000 in annual wellness spending per person.

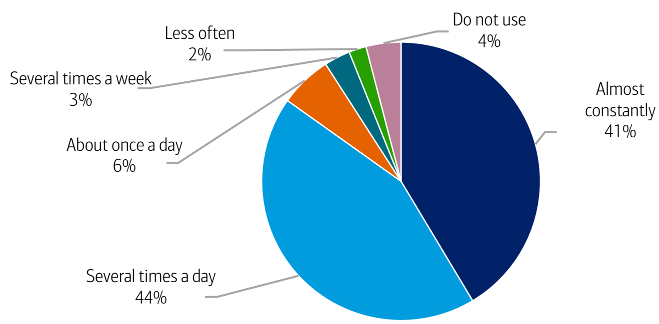
Tech is eating our health

Wellness, a multidimensional ecosystem aimed to improve health and quality of life, is not a new concept. However, as the digital revolution and ubiquitous connectivity have transformed our lives, so can the world of wellness. While technology delivers benefits and drives progress, it also introduces physical, social and psychological health risks, according to BofA Global Research. Constant connectivity and AI-driven tools have increased dependence on digital devices, with screen time replacing some in-person interaction. This shift has brought growing health considerations: prolonged screen exposure can contribute to poor posture, eye strain and obesity, while tech-related health issues – including loneliness, obesity, myopia, anxiety and depression – are estimated to cost the global economy at least $7 trillion annually, or about 6% of global GDP.

A tech-ubiquitous world

Nearly 6.12 billion people – almost three quarters of the global population (c.73.8%) – are now online.[1] On average, adult internet users spend six hours and 38 minutes online per day,[2] and four out of every 10 US adults say they're online almost constantly (Exhibit 1).

Exhibit 1: Roughly four in 10 say they're almost constantly online
% of adults who say they use the internet by frequency



Source: Pew Research Center[3], BofA Global Research

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[1] Data Reportal. (2026, April). *Digital Around the World.*
[2] Kemp, S. (2025, February 5). *Digital 2025: Global Overview Report*. Data Reportal.
[3] Pew Research Center. (2026, January 8). *Daily internet use is the norm for US adults, and about 4 in 10 say they're almost constantly online.* https://www.pewresearch.org/chart/daily-internet-use-is-the-norm-for-u-s-adults-and-about-4-in-10-say-theyre-almost-constantly-online/.

Total time a person spends on social media equates to 32 round trips to the moon

Further illustrating our scale of digital engagement, people now spend roughly five times more time on social media each day than socializing. At an average of 141 minutes per user per day, global social media use totals about 755 billion minutes daily.[4] Over a year, that amounts to more than 275 trillion minutes – or roughly 525 million years of collective human time.

Consumers increasingly engage with AI

Since ChatGPT's release in November 2022, generative AI adoption has accelerated rapidly. One global survey conducted by KPMG and the University of Melbourne found that of 48,340 respondents, 66% report using AI on a regular basis – for work, personal or academic reasons.[5]

Additionally, more consumers are beginning to pay for AI services, upgrading beyond free versions. Our recent publication, Not quite mAInstream: A consumer AI profile, notes that while only 3% of Bank of America households currently pay for AI services, that share is up 38% relative to the 2024 average.

Excessive tech use contributes to physical, social and psychological health risks

Excessive technology use can strain physical health in several ways. Poor posture can lead to discomfort such as neck, shoulder and back pain. Extended screen time tires the eyes and can interfere with sleep patterns. Screen-heavy, sedentary lifestyles also increase obesity risks;[6] notably, people now spend eight times more time online than being physically active.

Meanwhile, social and psychological risks are also rising. When time spent with technology displaces in-person social connection, feelings of isolation can intensify. In fact, those who report using social media for more than two hours per day are roughly twice as likely to report an increased perception of social isolation as those who use social media for less than 30 minutes per day.[7] And while loneliness may not be a formal medical diagnosis, it is just as widespread as major health concerns including obesity, smoking and diabetes.

At the same time, constant connectivity can cause digital fatigue. The never-ending flow of notifications, updates and information can create a sense of overload, making it harder to disconnect and increasing the risk of anxiety or persistent connectivity strain. Reflecting this dependence, about 44% of US teens report feeling anxious without their phones.[8]

Physical and mental health are interconnected

Each dimension of our health is mutually interdependent, according to BofA Global Research, meaning stress in one area often spills into another. Poor physical health, for example, can undermine mental wellbeing: the United Kingdom's National Health Service (NHS) finds that long-term or chronic conditions are associated with higher rates of anxiety, social isolation and sleep problems.

The reverse is also true. Mental health challenges are closely linked to physical illnesses like coronary heart disease, hypertension and diabetes,[9] as well as asthma, cancer and arthritis.[10] Around 40% of people with severe mental health symptoms also have long-term physical conditions, versus roughly 25% among those with few or no symptoms.[11]

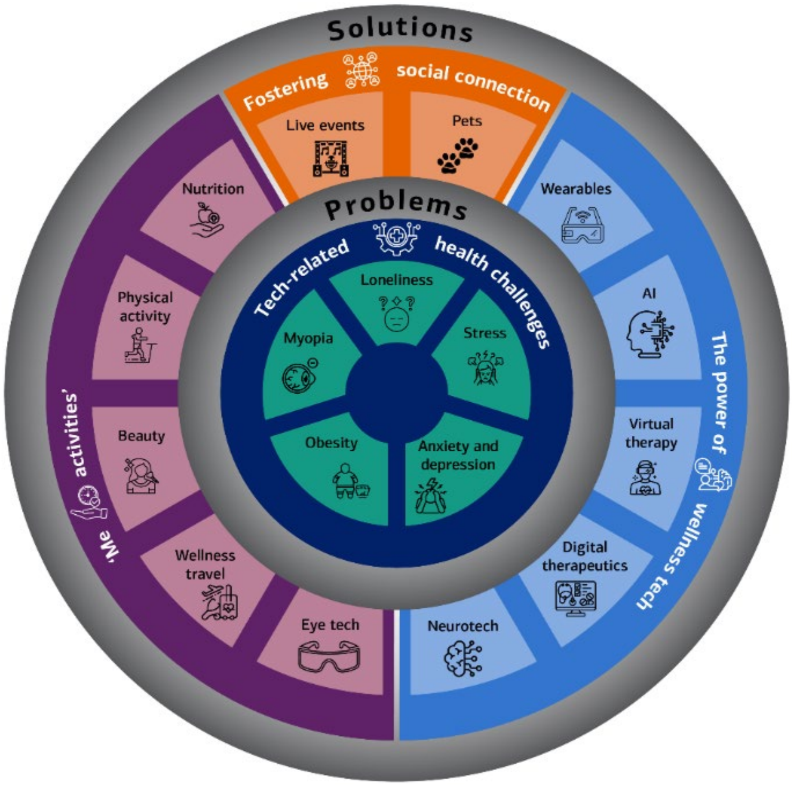
We need to take care of our physical, emotional and social wellbeing

According to BofA Global Research, wellness solutions are becoming more important to address tech-related health risks (Exhibit 2). Wellness is a multidimensional concept encompassing physical, emotional and social wellbeing – ranging from shared experiences and stress-reducing interactions with pets, to nutrition, physical activity and travel to reduce digital strain. While heavy screen use is linked to health risks, other technologies can support wellbeing, like virtual therapy and wearables. Together, all these forces underscore the rise of the wellness market.

[4] D., T. (2026, April 30). *Daily Time Spent on Social Networking by Internet Users Worldwide from 2012 to 2025*. Statista.
[5] Cardillo, A. (2025, October 27). *How many people use AI? (Latest 2025 Data)*. Exploding Topics.
[6] American College of Cardiology. (2026, March 24). *Excessive Screen Time Signals Health Risk for Young Adults*.
[7] Colditz, J.B, Lin, L.Y., Miller, E., Primack, B.A., Radovic, A.M, Rosen, D., Shensa, A., Sidani, J.E., & Whaite, E.O. (2018, July 1). *Social Media Use and Perceived Social Isolation Among Young Adults in the US*. PubMed Central.
[8] Anderson, M., Faverio, M., & Park, E. (2024, March 11). *How Teens and Parents Approach Screen Time*. Pew Research Center.
[9] Heinz, S.S., O'Brien, A.J., Parsons, M., & Walker, C. (2025, January 27). *Physical Health Views Among Individuals Experiencing Mental Illness: A Mixed-method Study of Self-reported Health and Contributing Factors*. PudMed Central.
[10] DerSarkissian, C. (2025, July 8). *How does mental health affect physical health?* WebMD.
[11] Mental Health Foundation. (n.d.). *People with Physical Health Conditions: Statistics*.

Exhibit 2: Solutions for tech-related health problems include fostering social connection, wellness tech and 'me activities'

Infographic illustrating tech-related health challenges and solutions



Source: BofA Global Research

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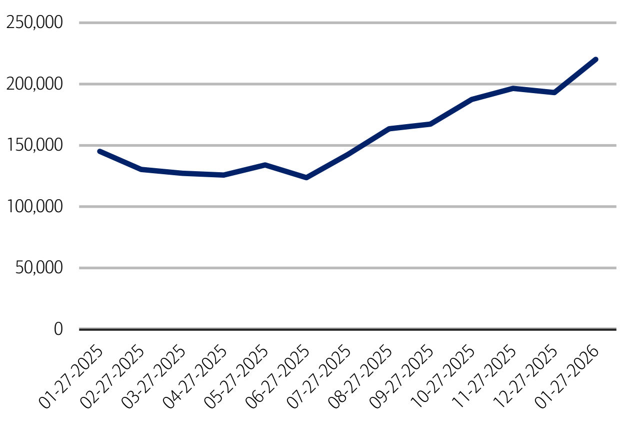
Furthermore, data from AlphaSense suggests that tech-related health risks (Exhibit 3) and wellness solutions (Exhibit 4) have become increasingly relevant in news coverage, trending upward since mid-2025.

Exhibit 3: Since January 2025, the number of news articles on tech-related health risks increased by 51%

Number of news articles published by date on tech-related health risks



Source: AlphaSense, BofA Data Analytics Team

Note: Number of news articles on keywords related to screen time, tech addiction, anxiety, depression, loneliness, social isolation, mental health, behavioral health, emotional health, stress, burnout, sedentary lifestyle, physical inactivity, obesity, metabolic health, myopia, digital eye strain, computer vision syndrome, digital overload and information overload. News articles refer to business news, general news, markets news, trade publications and thought leadership.

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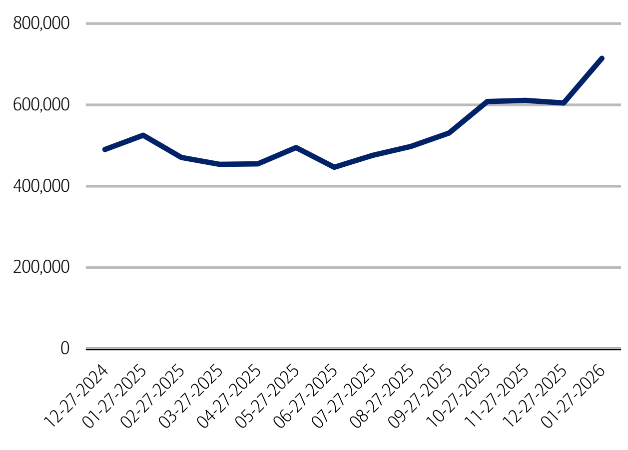
Exhibit 4: Since January 2025, the number of news articles relating to wellness solutions has increased by 36%

Number of news articles published by date on wellness solutions



Source: AlphaSense, BofA Data Analytics Team

Note: Number of news articles on keywords related to wellness, wellbeing, self-care, nutrition, diet, protein, weight management, obesity, weight loss, GLP-1, wearables, fitness, exercise, physical activity, occupational health, physical therapy, wellness travel, myopia management, virtual therapy, telemedicine, digital therapeutics, pets, live events, social connection, pharmacogenomic and precision medicine. News articles refer to business news, general news, markets news, trade publications and thought leadership.

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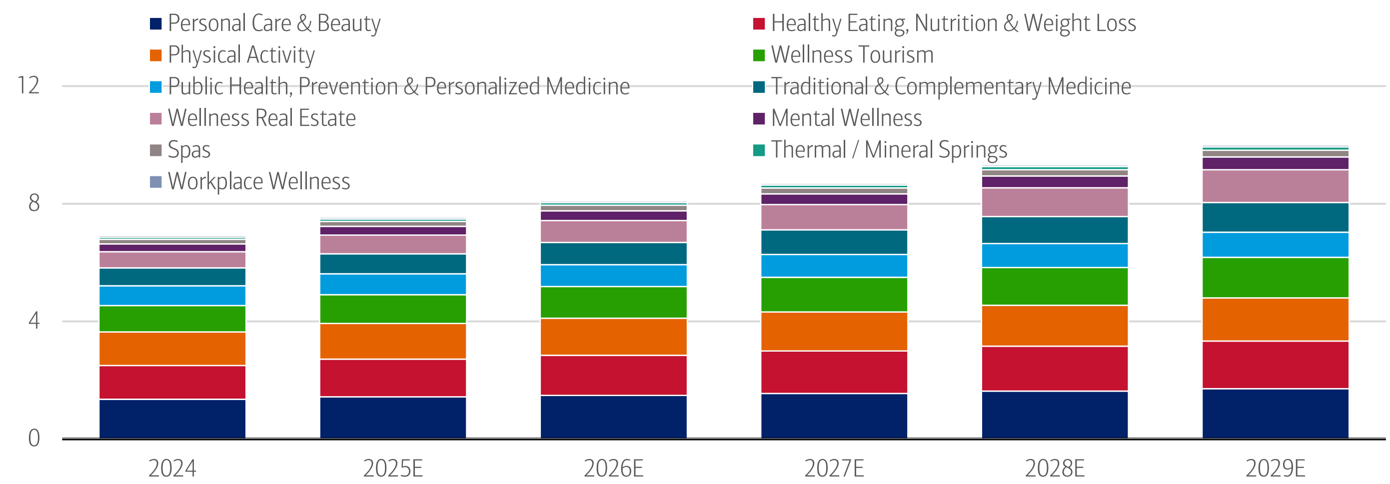
A $7 trillion global wellness economy

The US wellness economy is around $2 trillion, the largest in the world – and accounts for about 30% of the $7 trillion global wellness market (Exhibit 5).[12] With a population of around 340 million, that equates to approximately $6,000 per person annually spent on wellness. The US has ranked first since Global Wellness Institute (GWI) began measuring the sector in 2014 and continues to widen its lead.[13] Today, the US wellness industry is more than $1.1 trillion larger than China's – the second-largest market – and is more than six times larger than Germany's, which ranks third.[14]

Exhibit 5: The global wellness economy could be worth $10 trillion by 2029E, from $7 trillion in 2024 – growing at a 7.6% compound annual growth rate (CAGR)

Wellness economy market size by sector ($, trillions)



Source: Global Wellness Institute, BofA Global Research

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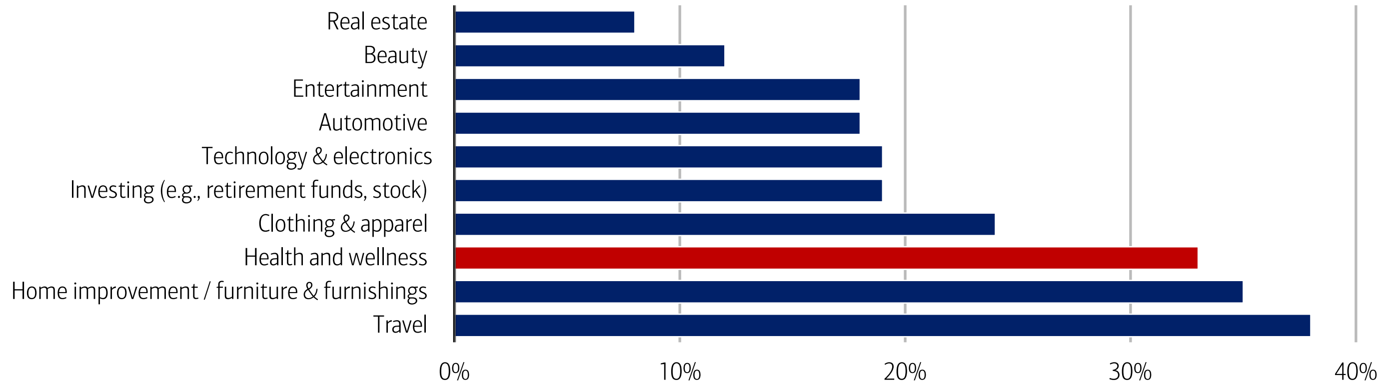
Consumers anticipate spending more on health & wellness in 2026

Underscoring the growth potential of the global wellness economy, a recent CivicScience survey shows that, among consumers intending to increase spending in 2026, roughly 32% plan to spend more on health and wellness (Exhibit 6). Additionally, about 36% plan to increase spending on travel and 20% on beauty – two categories that can support mental and physical wellbeing.

Exhibit 6: Nearly one in three people planning to increase their spending in 2026 intend to do so on health and wellness

% responses to "Which of the following categories are you likely to increase your spending on in 2026?"



Source: CivicScience

Note: 14, 348 responses from December 23, 2025, to May 14, 2026. See methodology.

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[12] Johnston, K. (2026). *The global wellness economy: United States data (2019-2024)*. Global Wellness Institute.

[13] Global Wellness Institute. (2025, March 4). *New research shows the US wellness economy – valued at $2 trillion – now represents one-third of the entire global wellness economy.*

[14] Ibid.

Methodology

The CivicScience survey includes 14, 348 responses from December 23, 2025, to May 14, 2026. This is an ongoing survey.

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Disclosures

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Transformation

Wellness, part 2: Fostering social connection

03 June 2026

Key takeaways

- Social connection plays a critical role in wellbeing, with social bonding activities releasing oxytocin, reducing stress and supporting cognitive health. While often viewed as individualistic, social connection is also shaped by broader infrastructure – such as community spaces, programs and local policies – with post-pandemic data pointing to rising demand for connection.
- Live events and pet ownership are key drivers of social connection, helping reduce loneliness while supporting economic activity. At the same time, rising remote work has contributed to workplace loneliness, which carries significant productivity costs - underscoring the importance of fostering connection both inside and outside the workplace.
- In the first installment of our series, we explored the wellness economy and its growth potential. Now, we turn to a key solution to tech-driven health challenges: fostering social connection.

What's the science behind social wellness?

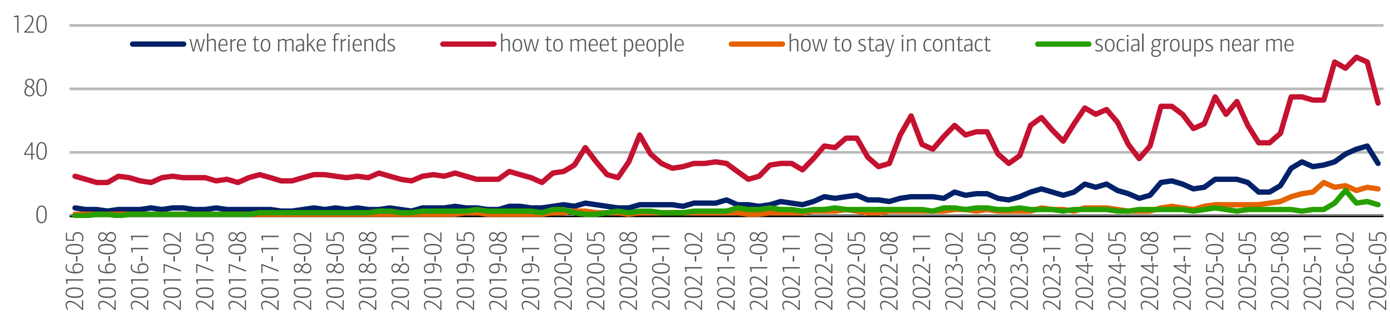
Oxytocin is a neurotransmitter released in the brain during social bonding activities – such as hugging, laughing or engaging in meaningful conversation. Social support can reduce cortisol (the body's stress hormone), which in turn lowers anxiety and supports immune function. And when social interaction occurs regularly, it also stimulates the brain, enhances memory and improves mental clarity.[1]

Social connection is influenced by social infrastructure

Social connection is often thought to be mostly driven by the individual (e.g., someone's personality) – but it's also shaped by the broader social infrastructure of communities. This includes physical assets like libraries and green spaces, as well as community programs (including volunteer organizations or other membership-based groups) and local policies around transportation and housing.[2]

Exhibit 1 shows that individuals are increasingly seeking ways to meet people and maintain social connections. Historically, this was not a major search-driven concern, with relatively low activity in the late 2010s. However, in the post-pandemic period, interest has steadily ticked up, with searches for "how to meet people" reaching peak levels in 2026.

Exhibit 1: People are increasingly interested in how to meet people and how to keep up social connections
Worldwide Google searches for the following terms



Source: Google Trends
Note: Data as of May 26, 2026. Numbers represent search interest relative to the highest point on the chart. 100 = peak popularity, 0 = not enough data.

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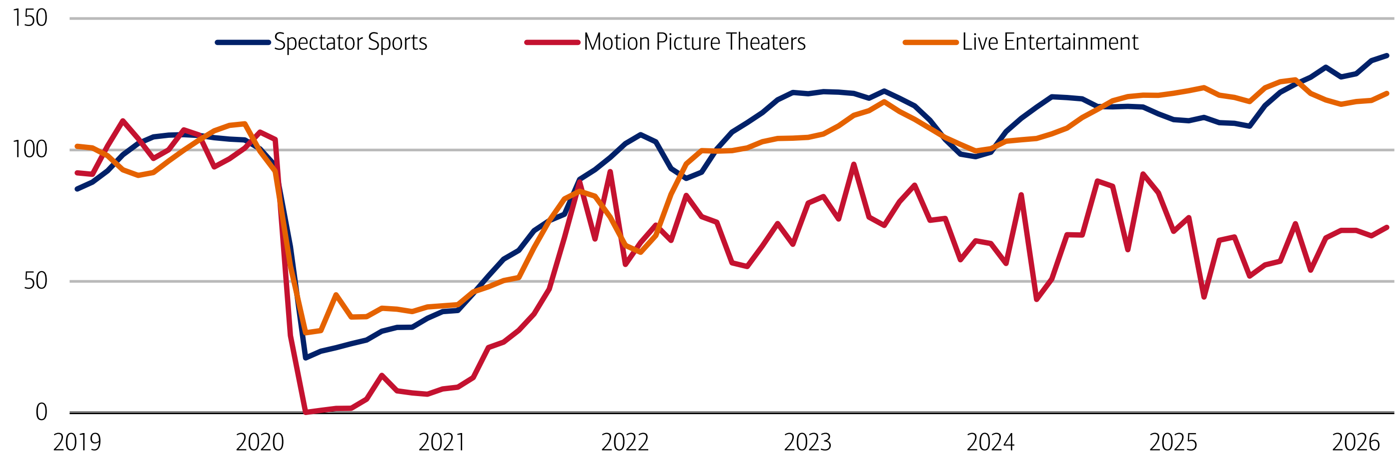
[1] Everyone Active.
[2] Office of the US Surgeon General. (2023). *Our Epidemic of Loneliness and Isolation: The US Surgeon General's Advisory on the Healing Effects of Social Connection and Community.*

Live events

In-person events that bring people together around shared experiences can play an important role in fostering social connection. Consumer demand for these experiences is evident in the data: the live entertainment market is projected to grow from $202.9 billion in 2025 to $270.3 billion by 2030 – a 5.9% compound annual growth rate (CAGR), according to Research and Markets.

Our own data has also highlighted the power of live entertainment – particularly sporting events – not only in bringing people together, but in supporting broader economic activity (read our publications: On the ball: How football fuels local spending and On the ball: Local economies score when sports kick off). Bureau of Economic Analysis data underscores this trend, showing strong engagement with event-related spending in recent years. As of March 2026, both spectator sports and live entertainment spending were above their 2019 averages, up roughly 36% and 21%, respectively (Exhibit 2).

Exhibit 2: Spending on live events continues to strengthen post-pandemic, especially for spectator sports
Real personal consumption expenditure on spectator amusements by type (monthly, seasonally adjusted, index 2019 = 100)



Source: Haver Analytics

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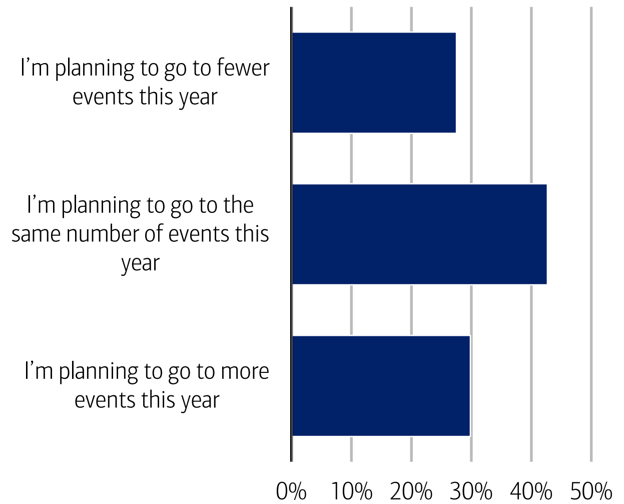
According to the 2026 Bank of America Summer Travel Outlook, nearly 30% of respondents plan to attend more live events this year compared to last, including concerts, sporting events and festivals (Exhibit 3). By generation, 91% of Gen Z plan to attend the same number of events or more – up from 85% in 2025 – while 80% of Millennials expect to maintain or increase attendance.

Exhibit 3: Nearly 73% of respondents plan to attend the same number or more events in 2026 than in 2025
% responses to the question, "Compared to 2025, are you planning to attend more concerts/sporting events/festivals this year?"



Source: 2026 Bank of America Summer Travel Outlook

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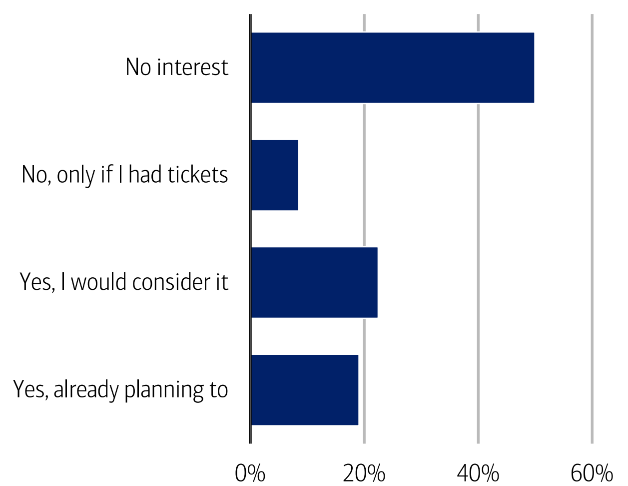
Exhibit 4: Some respondents would travel to a host city for the atmosphere, even without a FIFA World Cup 2026™ match ticket
% responses to question, "Would you travel to a FIFA World Cup 2026™ host city to experience the atmosphere, even if you don't have tickets to attend a match?"



Source: 2026 Bank of America Summer Travel Outlook

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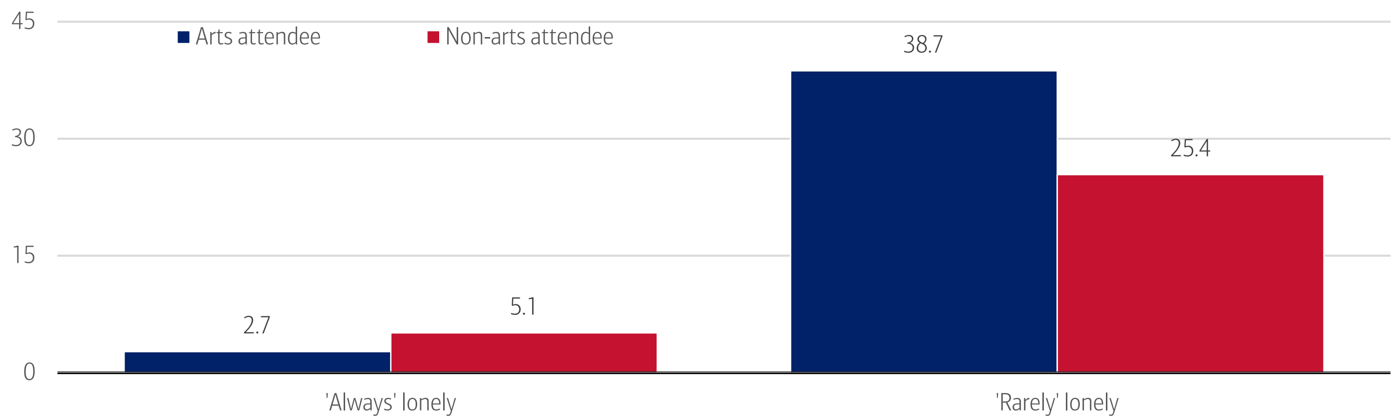
This sustained demand is reflected in consumers' willingness to make financial trade-offs to attend live events. Overall, 61% of Americans say they would adjust their spending to afford travel to a live entertainment event, rising to 88% among Gen Z. The most common trade-offs include cutting back on dining out, taking on extra work and using credit cards.

At a global level, this enthusiasm for live experiences is evident in large-scale events such as the FIFA World Cup 2026™, which is expected to draw nearly 6.5 million fans this summer – almost double the previous record. Historically, World Cup matches have attracted a cumulative 44 million spectators since 1930, underscoring the enduring appeal of live sports. Notably, around 40% of consumers say they would even travel to a host city this year without a match ticket, simply to experience the atmosphere (Exhibit 4). Check out our recent publication for more on this topic: [Summer Travel 2026: Resilient, but uneven](#).

While live events like concerts and sporting events create shared, high-energy moments of connection, arts-based experiences also play a role in fostering social connection. Between April and July 2024, roughly 25% of US adults attended at least one live, in-person performance and/or art exhibit in the previous month. And, according to the National Endowment for the Arts, US adults who attended live arts events were less likely than non-arts attendees to report feeling more acute levels of loneliness (Exhibit 5).

Exhibit 5: US adults who attended live arts events were less likely than non-arts attendees to report feeling loneliness
% of US adults who reported feeling lonely who attended or did not attend a live arts event



Source: National Endowment for the Arts, US Census Bureau's Household Pulse Survey from April to July 2024, BofA Global Research

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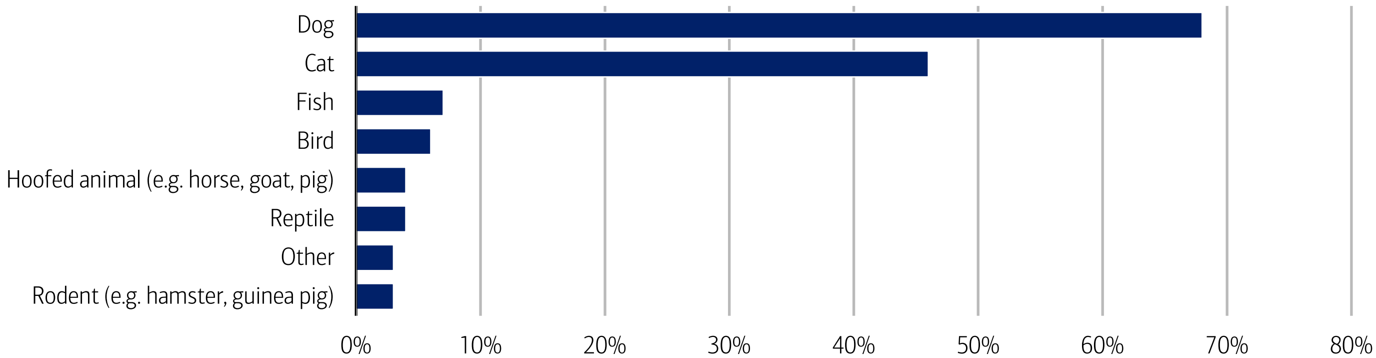
Pets

Pet ownership has surged over the past three decades: as of 2025, 71% of US households (or 95 million homes) own a pet – up from 56% in 1988.[3] According to a CivicScience survey, dogs remain the most popular companion – with 68% of respondents owning one (Exhibit 6). Cats aren't far behind at 46%. For more, read our recent pet publication: [The price of pet parenting has gone off leash](#).

Exhibit 6: Dogs and cats are the most popular pets in terms of ownership
What type of pet(s) do you currently have? (% of respondents with pets)



Source: CivicScience
Note: 78,445 responses from June 1, 2023 to May 8, 2026. See methodology.

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[3] American Pet Products Association. (2026, March 26). *2025 US Pet Industry Hits $158B, Set to Grow in 2026.*

This surge in ownership is also translating into significant spending. US pet expenditures reached $158 billion in 2025, according to the American Pet Products Association.[4] Globally, the pet care market – encompassing pet food and pet products – could grow from $207 billion in 2025 to $271 billion by 2030, representing a 5.5% compound annual growth rate (CAGR).[5]

Companionship pays (emotionally)

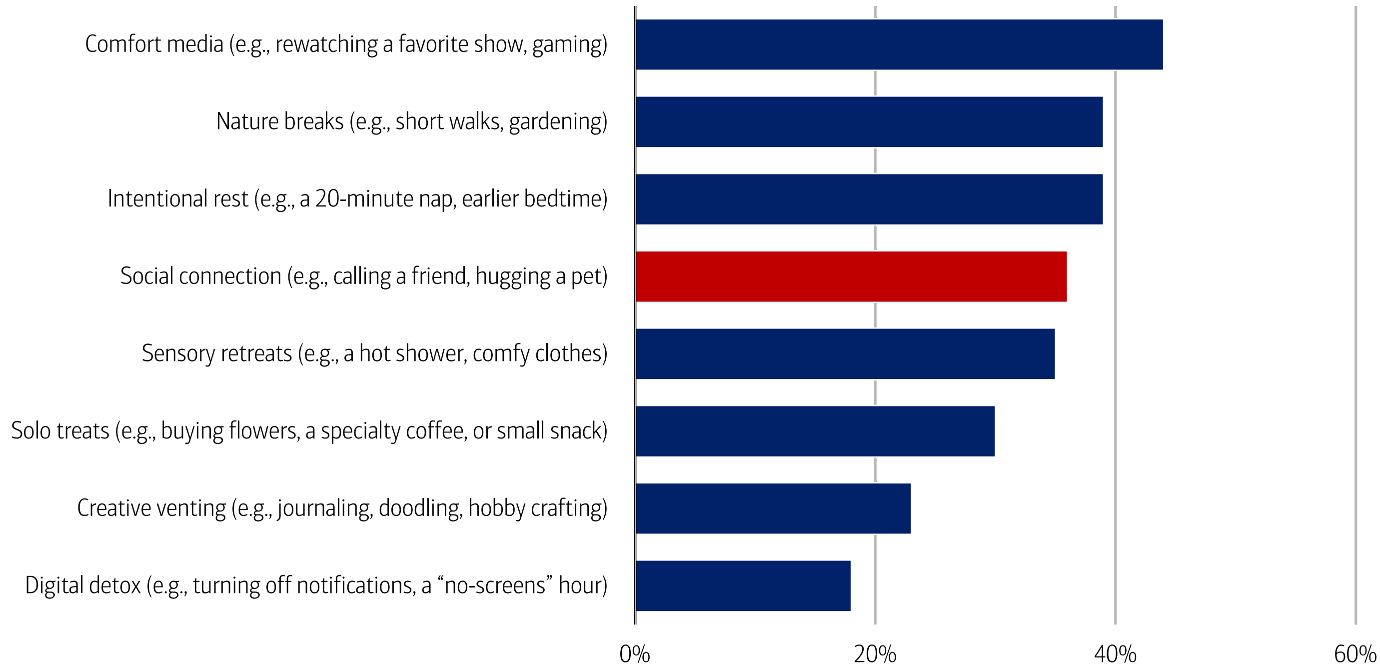
Studies have shown that spending time with pets can significantly increase oxytocin levels in humans and reduce cortisol.[6] Reflecting this, 97% of US pet owners consider their pets part of the family and 51% say their pets are as much a part of their family as human members, according to a Pew Research Center survey.[7]

For many, pets are also a primary source of stress relief. The American Heart Association finds that 95% of pet owners rely on their pet to manage stress.[8] The most popular ways pets help people are cuddling (68%), laughter (67%) or helping them feel less lonely (61%).[9]

Recent CivicScience survey data underscores pets' role in everyday wellbeing (Exhibit 7). When asked how they reset after a bad day, 36% of respondents cited social connection – including hugging a pet. Interestingly, 18% reported engaging in a digital detox – turning off notifications and reducing screen time (a theme we explored [Wellness, part 1: The rise of the wellness economy]).

Exhibit 7: 36% of respondents seek out social connection – including hugging their pets – after a bad day

% response to the question "Which of these 'small wins' do you use to reset your mental health or treat yourself after a bad day (select all that apply)?"



Source: CivicScience

Note: 8,672 responses from January 20, 2026 to May 18, 2026. See methodology.

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Teamwork and collaboration in the workplace

In 2024, a Gallup poll found that one in five employees worldwide feel lonely, rising to 27% among remote workers.[10] The share of employees working remotely (all or most of the time) increased from 20% in 2020 to 28% in 2023[11] and is set to grow

[4] American Pet Products Association. (2026, March 26). *2025 US Pet Industry Hits $158B, Set to Grow in 2026.*

[5] Puri, S. (2026, September 3). *Pricing pressure, premiumization and pet health: Trends shaping global pet care in 2025.* Euromonitor International.

[6] Beetz, A., Julius, H., Kotrschal, K., & Uvnas-Moberg, K. (2012, July 8). *Psychosocial and Psychophysiological Effects of Human-Animal Interactions: The Possible Role of Oxytocin.* Frontiers in Psychology.

[7] Brown, A. (2023, July 7). *About half of US pet owners say their pets are as much a part of their family as a human member.* Pew Research Center.

[8] American Heart Association. (2022, June 20). *New survey: 95% of pet parents rely on their pet for stress relief.*

[9] Ibid.

[10] Pendell, R. (2025, May 8). *The Remote Work Paradox: Higher Engagement, Lower Wellbeing.* Gallup.

[11] Slotta, D. (2025, December 17). *Work from home: remote & hybrid work – Statistics & Facts.* Statista.

further, with 73 million digital roles already able to be performed remotely worldwide – a figure that could reach 92 million by 2030.[12]

The implications are significant: employee disengagement costs the global economy roughly $9.6 trillion in lost productivity each year, equivalent to about 9% of world GDP.[13]

That said, there are ways to address workplace loneliness. Employers can foster a more positive, inclusive culture, recognize employee contributions, and offer mental health and wellbeing resources (Exhibit 8).

Exhibit 8: Solutions include a positive workplace culture, social interactions, flexible working and access to wellbeing resources
Infographic illustrating ways to tackle loneliness at work

Source: People Insight, BofA Global Research

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[12] Masterson, V. (2024, January 9). *More and more jobs can be done from anywhere. What does that mean for workers?* World Economic Forum.
[13] Gallup. (2026). *State of the Global Workplace.*

Methodology

Exhibit 6 is a CivicScience survey that includes 78,445 responses from June 1, 2023 to May 8, 2026. This is an ongoing survey.

Exhibit 7 is a CivicScience survey that includes 8,672 responses from January 20, 2026 to May 18, 2026. This is an ongoing survey.

The Bank of America 2026 Summer Travel Outlook survey was conducted online between March 26 and April 3. The survey consisted of 2,004 respondents throughout the U.S. Respondents in the study were age 18+ and were representative of the composition of the U.S. Census for age, gender, household income and Census region. Lower, middle and higher household income cuts in the survey were based on quantitative estimates of annual household income before taxes. They are defined as follows: lower-income households: $0-$66,000; middle-income households: $66,001-$130,000; higher-income households: >$130,000.

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Economy

The Institute Employment Report: May 2026

04 June 2026

Key takeaways

- Payroll growth picked up again in May, according to Bank of America customer deposit account data, suggesting the labor market remains resilient and broadly healthy. It appears much of this strength is being driven by gains in lower-income jobs.
- Consistent with improvements in the labor market, unemployment payments into Bank of America customer deposit accounts continues to show slowing growth.
- Wage growth for lower- and middle-income households is recovering but still lags higher-income earners, despite some narrowing in the gap. Lower- and middle-income after-tax wage growth rose to 3.1% and 3.5% year-over-year (YoY), respectively, in May, while higher-income wage growth eased to 5.6% YoY.

May continued to see solid payrolls growth

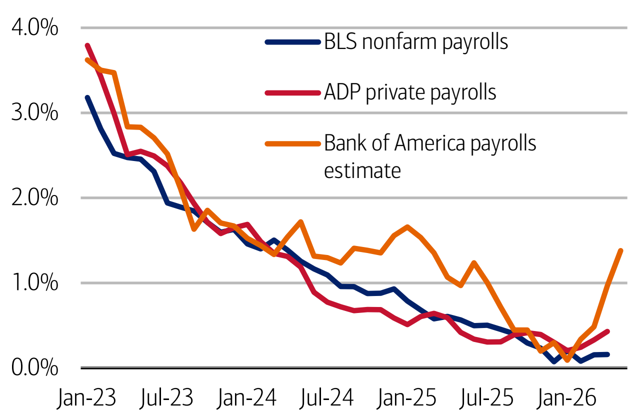
We use Bank of America consumer deposit data to estimate a payrolls series by looking at how the number of customer accounts receiving a paycheck is changing (see Methodology). This data suggests payrolls grew briskly in May. And the acceleration in jobs growth since the start of the year appears to be concentrated in lower-income roles.

This data can be fairly noisy, partly due to seasonal variation and variation in pay periods. This month's data also incorporates some revisions to our data, to better identify payments aligned with job-related income. While these revisions lower recent year-over-year (YoY) growth rates, they do not change the directional story of rising jobs growth in 2026.

Looking at the three-month moving average, Exhibit 1 shows our jobs growth measure rose to 1.4% year-over-year (YoY) in May, up from 1.0% in April. This is broadly in line with the rates seen in late 2024/early 2025 – a period of healthy payrolls growth according to the Bureau of Labor Statistics' (BLS) measure.

Exhibit 1: Bank of America account data indicates payrolls growth continued into May

Payroll estimates from Bank of America customer deposit account data (three-month moving average, % YoY), the Bureau of Labor Statistics (BLS) and Automatic Data Processing (ADP) (monthly, YoY)



Source: Bank of America internal data, Haver Analytics

Note: BLS and ADP data are seasonally adjusted, Bank of America data is not seasonally adjusted.

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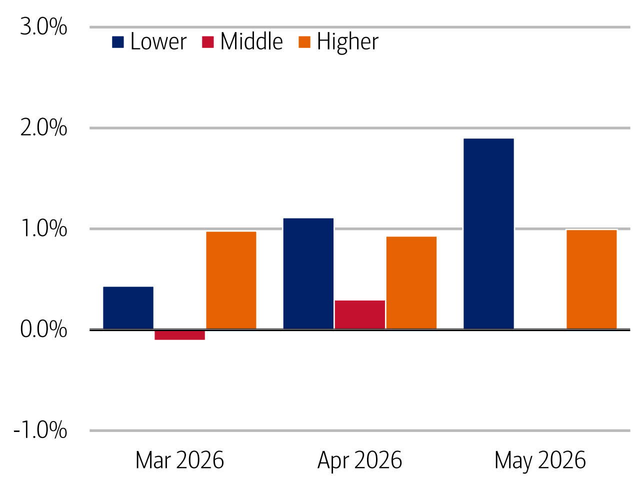
Exhibit 2: The acceleration in payrolls growth appears to be concentrated primarily in lower-income jobs

Payroll estimates from Bank of America customer deposit account data by household income tercile (three-month moving average, % YoY)



Source: Bank of America internal data

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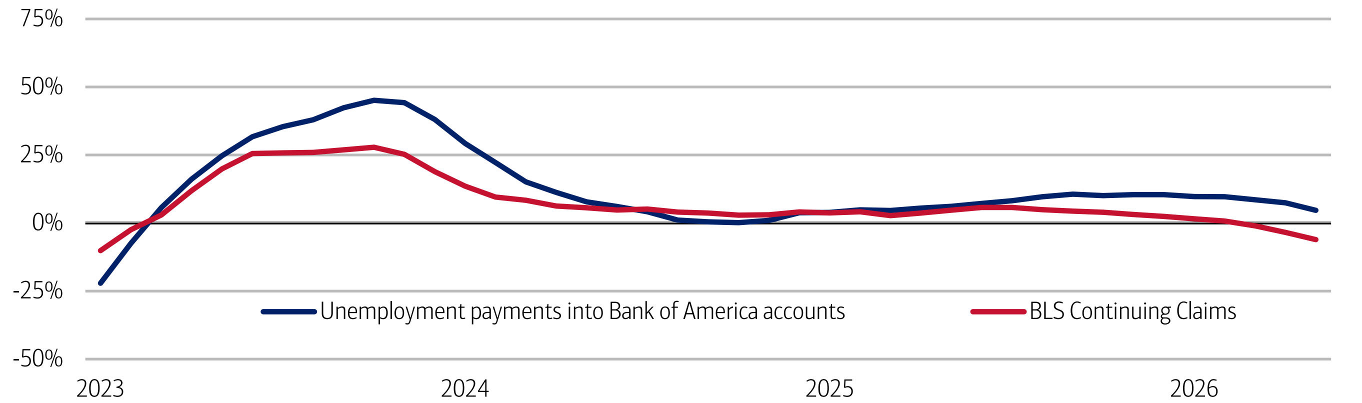
Interestingly, when we split our estimate of jobs growth by household income terciles, we find the acceleration in jobs growth in our data appears to be driven by lower-income payrolls (Exhibit 2).

Consistent with continued improvements in jobs growth, Bank of America data on unemployment payments into customer accounts continues to show easing growth (Exhibit 3).

Exhibit 3: Unemployment payment growth continued to decline in May

Number of households receiving unemployment payments (three-month moving average, YoY%, not seasonally adjusted (NSA)) and Continuing Claims (three-month moving average, YoY%, seasonally adjusted (SA))



Source: Bank of America internal data, Bloomberg

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Signs of recovery in lower- and middle-income wage growth?

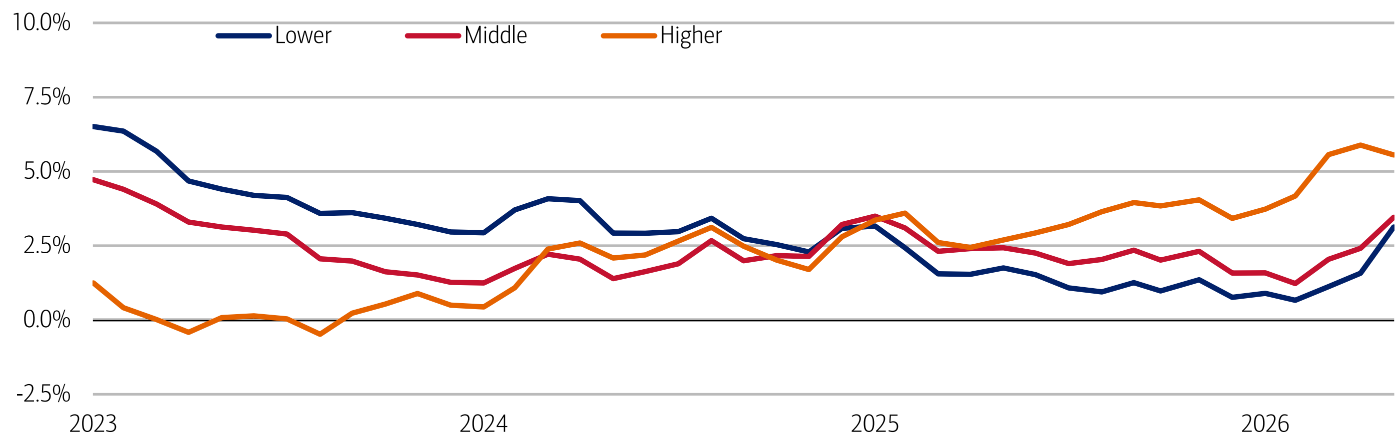
While there continues to be a large gap between higher-income households' after-tax wage growth and that of lower- and middle-income households in Bank of America data, there are signs of a recovery for the latter in May.

In May, higher-income households saw after-tax wage growth of 5.6% YoY – down from 5.9% YoY in April, but still a relatively high growth rate compared to the previous three years (Exhibit 4). The YoY growth rates for lower- and middle-income households are considerably lower, but both are continuing to recover. In May, lower-income households' after-tax wage growth rose to 3.1%, the highest rate since January 2025, while middle-income households' wage growth rose to 3.5% YoY. While still significant, the gap between higher- and lower-income households wage growth was the lowest it has been since July 2025.

What is driving this rebound in lower- and middle-income wage growth? In our view, some recovery is consistent with the stronger jobs growth we have seen in recent months, alongside an increase in job openings seen in the recent Job Openings and Labor Turnover Survey (JOLTS) report from the BLS. But we view the uptick in wage growth cautiously – sometimes changes in pay growth can reflect differences in the number of pay periods, for example. So we need to see additional data to confirm the trend.

Exhibit 4: Lower- and middle-income households saw after-tax wage growth rise to 3.1% and 3.5% YoY, respectively, in May, narrowing but not closing the gap with higher-income households

After-tax wage and salary growth by household income terciles, based on Bank of America aggregated consumer deposit account data (three-month moving average, YoY%, SA)



Source: Bank of America internal data

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Methodology

Selected Bank of America transaction data is used to inform the macroeconomic views expressed in this report and should be considered in the context of other economic indicators and publicly available information. In certain instances, the data may provide directional and/or predictive value. The data used is not comprehensive; it is based on **aggregated and anonymized** selections of Bank of America data and may reflect a degree of selection bias and limitations on the data available.

Any payments data represents aggregated spend from US Retail, Preferred, Small Business and Wealth Management clients with a deposit account or credit card. Aggregated spend include total credit card, debit card, ACH, wires, bill pay, business/peer-to-peer, cash, and checks.

Any **Small Business** payments data represents aggregate spend from Small Business clients with a deposit account or a Small Business credit card. Payroll payments data include channels such as ACH (automated clearing house), bill pay, checks and wire. Bank of America per Small Business client data represents activity spending from active Small Business clients with a deposit account or a Small Business credit card and at least one transaction in each month. Small businesses in this report include business clients within Bank of America and generally defined as under $5mm in annual sales revenue.

Unless otherwise stated, data is not adjusted for seasonality, processing days or portfolio changes, and may be subject to periodic revisions.

The differences between the total and per household card spending growth rate (if discussed) can be explained by the following reasons:

1. Overall total card spending growth is partially boosted by the growth in the number of active cardholders in our sample. This could be due to an increasing customer base or inactive customers using their cards more frequently.

2. Per household card spending growth only looks at households that complete at least five transactions with Bank of America cards in the month. Per household spending growth isolates impacts from a changing sample size, which could be unrelated to underlying economic momentum, and potential spending volatility from less active users.

3. Overall total card spending includes small business card spending while per household card spending does not.

4. Differences due to using processing dates (total card spending) versus transaction date (per household card spending).

5. Other differences including household formations due to young adults moving in and out of their parent's houses during COVID.

Any household consumer deposit data based on Bank of America internal data is derived by anonymizing and aggregating data from Bank of America consumer deposit accounts in the US and analyzing that data at a highly aggregated level. Whenever median household savings and checking balances are quoted, the data is based on a fixed cohort of households that had a consumer deposit account (checking and/or savings account) for all months from January 2019 through the most current month of data shown.

Bank of America aggregated credit/debit card spending per household includes spending from active US households only. Only consumer card holders making a minimum of five transactions a month are included in the dataset. Spending from corporate cards are excluded. Data regarding merchants who receive payments are identified and classified by the Merchant Categorization Code (MCC) defined by financial services companies. The data are mapped using proprietary methods from the MCCs to the North American Industry Classification System (NAICS), which is also used by the Census Bureau, in order to classify spending data by subsector. Spending data may also be classified by other proprietary methods not using MCCs.

We consider a measure of services necessity spending that includes but is not limited to childcare, rent, insurance, public transportation, and tax payments. Discretionary services includes but is not limited to charitable donations, leisure travel, entertainment, and professional/consumer services. Holiday spending is defined as items in which spending in the November-December period is usually at least 20% of total annual spending on the category.

For analysis looking at higher value transactions (including durables), we consider a value per transaction threshold estimated with reference to the top 30% of transactions by value in 2024. The share of higher value transactions is then the number of transactions above this threshold as a percentage of total transactions over time.

Lower, middle and higher household income cuts in Bank of America credit and debit card spending per household, and consumer deposit account data are based on quantitative estimates of each households' income. These quantitative estimates are bucketed according to terciles, with a third of households placed in each tercile periodically. The lowest tercile represents 'lower income', the middle tercile represents 'middle income' and the highest tercile 'higher income'. The income thresholds between these terciles will move over time, reflecting any number of factors that impact income, including general wage inflation, changes in social security payments and individual households' income. The income and tercile in which a household is categorised are periodically re-assessed.

Generations, if discussed, are defined as follows:

1. Gen Z, born after 1995
2. Younger Millennials: born between 1989-1995
3. Older Millennials: born between 1978-1988
4. Gen Xers: born between 1965-1977
5. Baby Boomer: 1946-1964
6. Traditionalists: pre-1946

Any reference to card spending per household on gasoline includes all purchases at gasoline stations and might include purchases of non-gas items.

Estimate of payrolls growth from Bank of America internal data is based on the change in customer accounts receiving a paycheck in the month. An adjustment is made for the difference between overall population growth and customer account growth.

An estimate of bonus growth from Bank of America deposit data is calculated by looking at customers who have received an inbound ACH payroll transaction in the last two years. From this sample an estimate of bonuses is derived by looking for payroll transactions which are over 50% higher than the median regular payroll payments received by the customer. Of these payments only those that were received around the same time in each of the last two years are selected.

Additional information about the methodology used to aggregate the data is available upon request.

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Disclosures

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Sustainability

Data center construction creates a resource shock

02 June 2026

Key takeaways

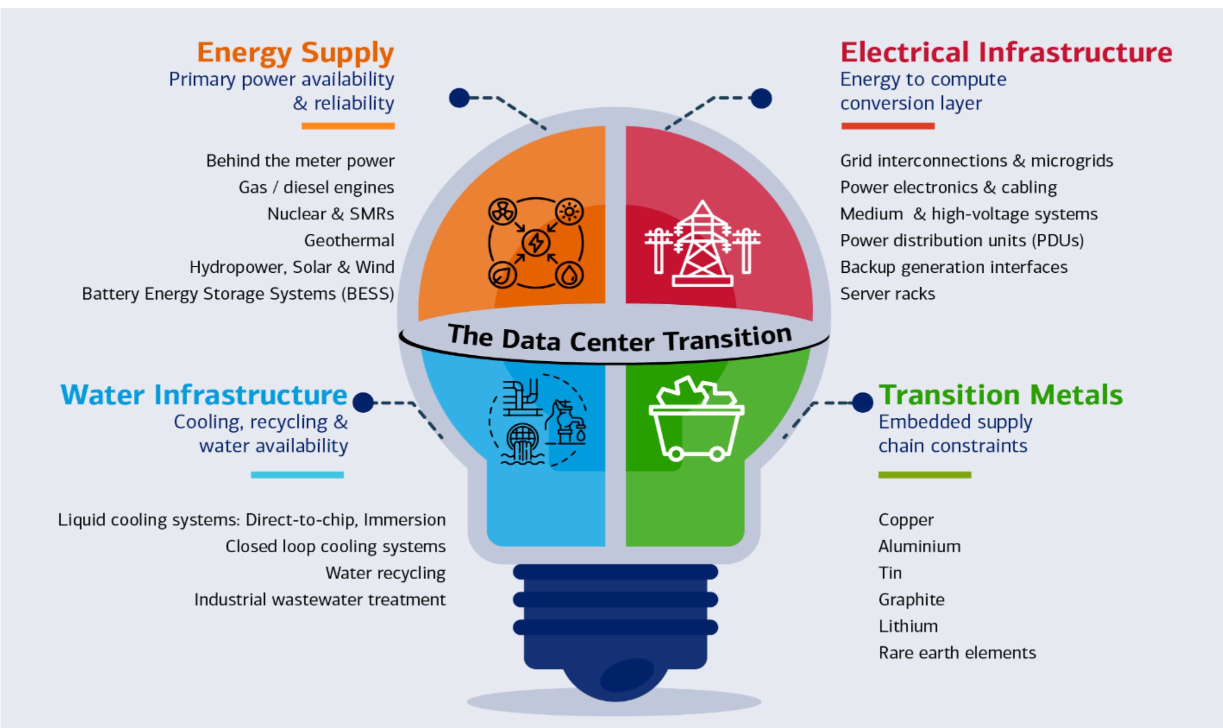
- AI's water footprint is a substantial resource drain. Up to 75% of a data center's total water use happens off-site, largely through electricity generation rather than on-campus cooling. At scale, even small AI interactions add up: millions of liters of water per day are indirectly tied to routine AI use, according to BofA Global Research.
- Electricity demand from graphics processing unit-based servers is growing at roughly 30% per year, driven increasingly by AI inference. This shift stresses local grids, where the challenge is no longer total energy supply but the ability to deliver firm, continuous power at the right location.
- Each incremental megawatt of data center capacity embeds roughly 60-75 tons of metals, particularly copper, according to BofA Global Research. As data centers scale, they steadily pull on water, power, and materials that local infrastructure was never designed to supply at this pace, intensifying pressure on already-constrained regions.

Breaking down a data center's value chain

AI is turning data centers into some of the world's fastest-growing industrial water users. Every graphics processing unit (GPU) cycle adds heat; every watt of power demands cooling and cooling demands water.

Exhibit 1: AI data center growth is increasingly shaped by energy supply, electrical and water infrastructure and critical transition metals

Infographic depicting the AI data center value chain



Source: BofA Global Research

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When compute heats up, the water runs out

AI is becoming one of the fastest-growing industrial users of water, and most of that demand sits outside the data center itself.

For a typical data center, only around a quarter of total water use is direct (Scope 1): cooling towers, humidification and other visible thermal systems. Most water is used off-site, split between electricity generation (Scope 2): particularly fossil and nuclear power plants and upstream hardware manufacturing, including semiconductor and chip fabrication. In aggregate, off-site water use can account for 75% of a data center's total water footprint.

Using a 100-word AI prompt? This will need a half-liter of water. Scale that up, and mid-sized data centers are drinking 1-2 million (mn) liters a day. By 2030, the data center annual water tab could hit 1.2 trillion (tn) liters – equivalent to New York City's entire annual water use.

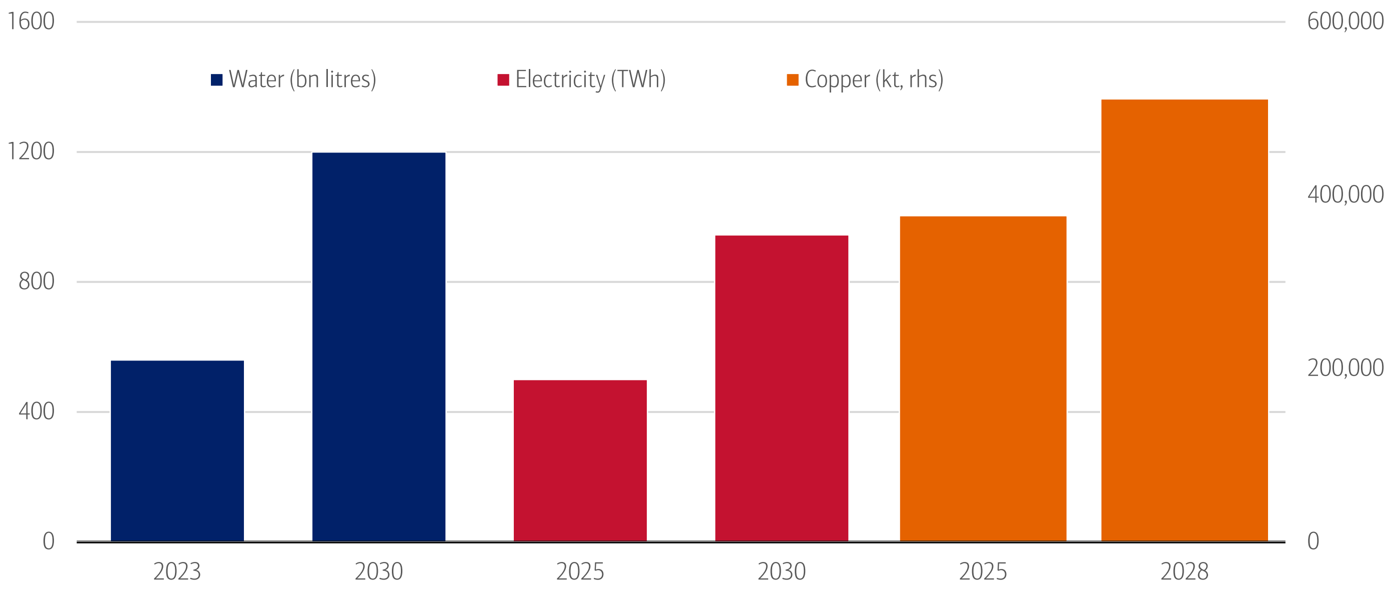
AI value chain is getting thirstier: Far larger with hidden footprint

Cooling loads are exploding as GPUs run hotter and denser, forcing operators toward increasingly water-hungry thermal systems. As power demand for AI soars, most grids still rely on water-intensive fossil and nuclear plants, meaning every new megawatt (MW) of AI compute carries significant hidden water use. Upstream manufacturing is an extreme user: semiconductor fabrication plants are water behemoths, consuming roughly 10 million gallons of ultrapure water per day to produce advanced chips.

The net result is that AI's water footprint is not just growing, but increasingly system-wide, opaque and geographically misaligned with where risks are measured or managed. For example, a data center can report an excellent Water Usage Effectiveness (WUE) – annual water consumption for humidification and cooling by the total energy consumption for information technology (IT) equipment – and still consume vast amounts if it is plugged into a water-intensive grid. In most markets, grid water intensity, not cooling system design, is the dominant driver of operational water risk.

Exhibit 2: Global data center capacity is expected to exceed 200GW (gigawatt) by 2030 and triple by 2035 versus 2024, driving a near doubling of electricity and water demand by 2030

Data center resource consumption by 2030



Source: International Energy Agency, Bloomberg NEF, BofA Global Research

Note: bn = billion; TWh = terawatts per hour; kt = kilotons; rhs = right-hand side

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Cooling sits at the center of AI's water challenge

From an operator's perspective, cooling is the only lever of immediate, engineering-level control over water use. As a result, cooling strategies are evolving fast. Data-center cooling operates at both the server and building level and next-generation cooling is shaping one of the biggest capex waves in hyperscale infrastructure. Closed-loop liquid cooling, immersion cooling and free-air cooling are emerging as competitive differentiators and a fast-growing capex category for original equipment manufacturers (OEMs).

Next-generation designs are fueling a new wave of capex

Data centers rely on a mix of cooling systems – from room-level air conditioners and chillers to liquid systems that cool chips directly – to manage the rising heat generated by powerful AI chips (Exhibit 3). Traditional air-based cooling often uses evaporation, which consumes significant water, while liquid-based cooling systems circulate water in closed loops and are far more water-efficient.

As AI models become more advanced, the chips that run them use more power and generate more heat. Servers are also being packed more tightly, concentrating that heat in smaller spaces. Effective cooling is therefore essential to keep systems running smoothly; without it, excess heat can degrade performance, shorten equipment lifespan and increase the risk of failures.

Inside the server rack, liquid cooling is emerging as the preferred solution. BofA technology analysts expect direct liquid cooling – where liquid removes heat directly from the chip – to remain the dominant approach, as it can handle high heat loads reliably within limited space.

At the building level, data centers typically rely on one of two methods to release heat: air-based cooling, which uses little water but more electricity, or evaporative cooling, which reduces energy use but requires much more water. Local climate conditions often determine which option is practical.

Exhibit 3: Data centers rely on a mix of cooling systems – from room-level air conditioners and immersion cooling to liquid systems

Types of server cooling system

Factor	Air Cooling	Evaporative Air Cooling	Direct-to-Chip Liquid Cooling	Rear-Door Heat Exchangers	Immersion Cooling
How it works	Air conditioning units use chilled air to cool the facility. Cold air circulates to the racks, while hot air returns to the air handling units for cooling and recirculation.	Air cooling where incoming air temperature is reduced via water evaporation (adiabatic process) before or during supply to IT equipment.	Liquid coolant circulates through cold plates on CPUs/GPUs and returns to a cooling distribution unit in a closed loop.	Liquid coolant circulates through heat exchangers mounted on the rear of each rack.	Servers are submerged in dielectric fluid that absorbs heat and circulates through a heat exchanger.
Water use	Low (no evaporation).	Evaporative systems consume water by design.	Very low, small, fixed volume in a closed loop; water use depends on heat-rejection method.	Low as closed-loop coolant; minimal water use at building level.	Minimal as synthetic fluids replace water entirely.
Energy efficiency at higher rack densities	Lower efficiency at higher rack densities.	More energy-efficient than mechanical air cooling; attractive in power-constrained grids.	Liquids have orders-of-magnitude higher heat capacity than air, enabling much higher rack densities.	Moderately efficient, improves rack-level thermal performance.	Highest efficiency; supported at rack/system level.

Source: BofA Global Research

Note: This is not an exhaustive list of technologies and factors; it is provided for illustrative purposes only.

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The $58bn water infrastructure gap

Most US water utilities have yet to fully account for the water implications of AI-driven infrastructure growth. A 2025 survey of 680 US water-sector stakeholders points to significant planning gaps.

- **Limited visibility of data-center demand**: Only 14% of respondents reported direct water demand from data centers including AI facilities in their service areas, even as development expands into secondary and rural regions.
- **Little impact on reuse planning so far**: Around two-thirds of respondents reported no increase in demand for recycled or repurposed water linked to data centers or advanced manufacturing; just 12% replied in the affirmative, but that figure is likely to rise as reuse becomes more viable for cooling.
- **Water planning lagging infrastructure growth**: 54% of respondents have not incorporated data center and advanced manufacturing water needs into resource planning, while only 13% have done so.[1]

This lack of preparedness contrasts sharply with demand projections. Academic research estimates that US utilities may require an additional 697mn to 1.45bn gallons per day of peak water-supply capacity by 2030 to support AI-driven data-center cooling. The associated investment needed to expand water infrastructure is estimated at $10bn to $58bn, depending on the pace and concentration of data center growth.[2]

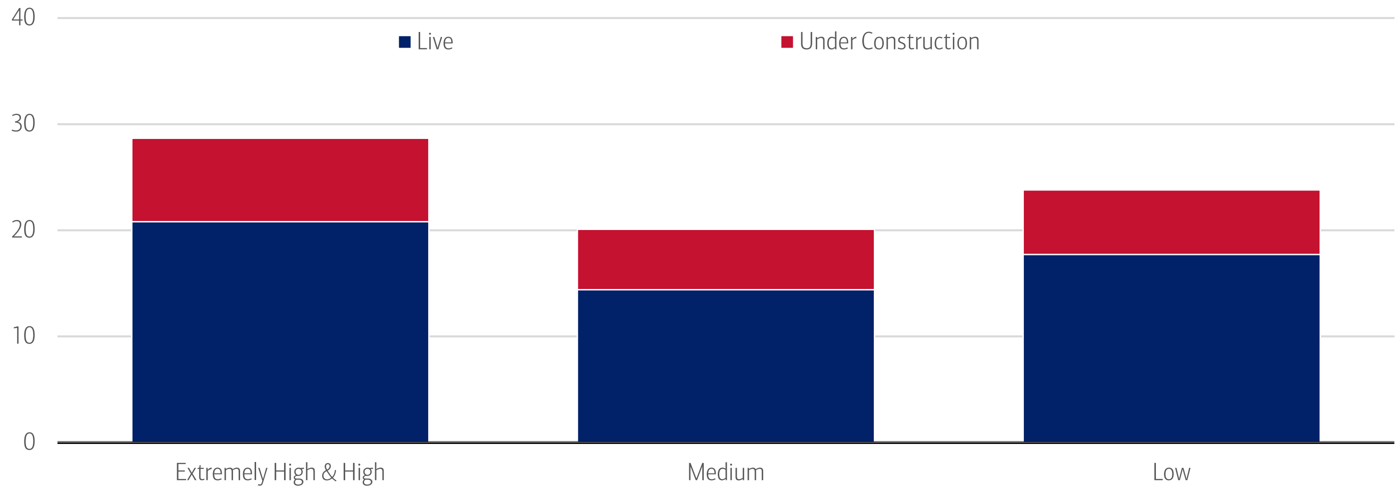
[1] Black & Veatch. (2025). Black & Veatch 2025 Water Report.
[2] Han, et. al. (2026, March 3). *Small Bottle, Big Pipe: Quantifying and Addressing the Impact of Data Centers on Public Water Systems.* University of California, Riverside.

Data centers are being built where the water isn't

Water availability is increasingly seen as an important factor in deciding where new AI computing capacity should be built. In practice, however, data centers continue to cluster in water-stressed regions, where access to power, available land and more permissive regulations often outweigh local water constraints, according to BofA Global Research.

Exhibit 4: Over 20GW of live global data center capacity is in extremely high or high water-stress regions

Total capacity by water stress level (in MW). Medium =10-20%, Low <10%



Source: Bloomberg, BofA Global Research

Note: Bloomberg's Data Centers and Water Stress Exposure Dataset maps global IT capacity to baseline and projected water stress levels.

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Adaptation needs: Wastewater and non-potable water solutions

Water impacts do not end at withdrawal. Alternative and non-potable water sources account for less than 5% of total water used by a typical data center.[3] This highlights how early the transition remains, even in regions exposed to water stress. Cooling systems fundamentally alter both the quantity and quality of water downstream, creating challenges that extend beyond freshwater scarcity.

In evaporative cooling systems, around 75-80% of withdrawn water is irreversibly consumed through evaporation. The remainder is discharged as blowdown wastewater, containing elevated dissolved solids, heat load and treatment chemicals such as biocides and corrosion inhibitors. Most municipal wastewater treatment plants were not designed to handle this industrial chemistry at hyperscale volumes, creating friction as data center density increases.

When combined with water-intensive electricity generation, this produces a water-energy-wastewater feedback loop that utilities are increasingly scrutinizing. As a result, wastewater handling is becoming as important as freshwater withdrawal in permitting decisions, with emerging requirements around pre-treatment, discharge quality and developer-funded infrastructure upgrades.

How cooling water is managed today and where efficiency gains lie

Most cooling systems recirculate water through cooling towers or heat exchangers until quality thresholds are reached. Some facilities discharge used water to municipal treatment plants or directly to receiving waters subject to temperature and chemical limits. Evaporation remains the dominant source of water loss, while poor maintenance can also lead to leakage, drift and overflow, creating hidden inefficiencies.

Advanced approaches including closed-loop cooling, wastewater recycling and rainwater harvesting can reduce freshwater demand by an estimated 50-70%, depending on climate and system design, but require rigorous treatment and monitoring.[4]

Policy signals heat up

In the United States, policy responses remain fragmented and largely local. According to BofA Global Public Policy, the White House is largely focused on accelerating data center development through permitting reform, infrastructure prioritization and support for global AI leadership. At the Congressional level, proposals reflect growing concern over ratepayer protection, environmental impacts and the need for greater transparency and accountability while prioritizing global competitiveness.

[3] Rosenfield, et. al. (2025, July 14). *Data centers and water.* Norton Rose Fulbright Project Finance Newswire.

[4] Ramboll and Grundfos. (2024, May 15). *Whitepaper: Water circularity can reduce water stress and boost EU industry.* Grundfos

In contrast, states and localities are advancing a wide range of approaches. Southern Nevada has prohibited evaporative cooling in new developments due to severe water stress, while California's proposed AB 93 would require businesses to disclose water and energy use and could link efficiency standards to operating licenses. Florida has also begun to publicly debate water use by AI data centers – some of which reportedly consume up to 500,000 gallons per day – prompting discussions around new water-management legislation, a signal that oversight is likely to intensify, according to BofA Global Research.

Policy is also beginning to condition incentives based on reclaimed-water use. The proposed Virginia Senate Bill 417 (2026 session) would require data-center operators receiving state infrastructure grants to progressively increase the share of reclaimed water used for cooling from 60% in 2027 to 100% by 2031 if enacted. A small number of large data center campuses already rely on reclaimed or treated wastewater for cooling. In parts of Georgia, multiple developments access treated municipal wastewater, reducing the reliance on potable supplies.

Not just a drain on water: The data center transition is electrical

By 2030, data centers could exceed 3% of global electricity demand. Electricity use from accelerated (GPU-based) servers is growing c.30% per year, versus c.9% for conventional central processing units (CPU) servers. Accelerated servers account for nearly half of the total increase in data-center electricity demand; conventional servers contribute c.20%, with the remaining c.30% coming from cooling and infrastructure.

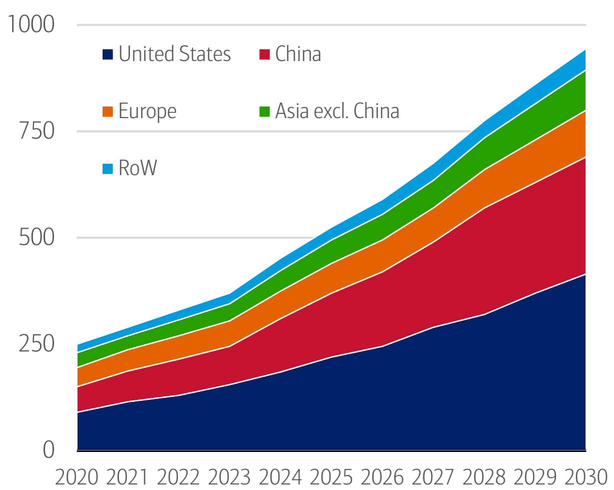
The key driver is no longer model training alone, but inference, especially reasoning-intensive and agentic workloads. As AI use expands from text into image, video and multimodal tasks, energy per task increases, utilization becomes near continuous, and workloads become harder to interrupt (read more on this in On the clock: Agentic AI in the workplace).

Higher power density, sustained load and infrastructure intensity

Cooling alone accounts for c.7% of electricity use in efficient hyperscale sites and >30% in less-efficient enterprise facilities. The US Department of Energy (DOE) explicitly identifies AI servers and cooling as key drivers of rising electricity demand.

The broader implication is that data center growth should not be analyzed as a single number in terawatt-hour (TWh); it should be analyzed as a stack: server power, cooling, UPS, batteries, conversion losses, backup generation and network reinforcement. This is where the next leg of value creation is building.

Exhibit 5: Data centers to consume close to 1,000TWh in 2030
Data center electricity consumption, 2020-2030, IEA base case



Source: International Energy Agency (IEA), BofA Global Research estimates
Note: RoW = rest of world.

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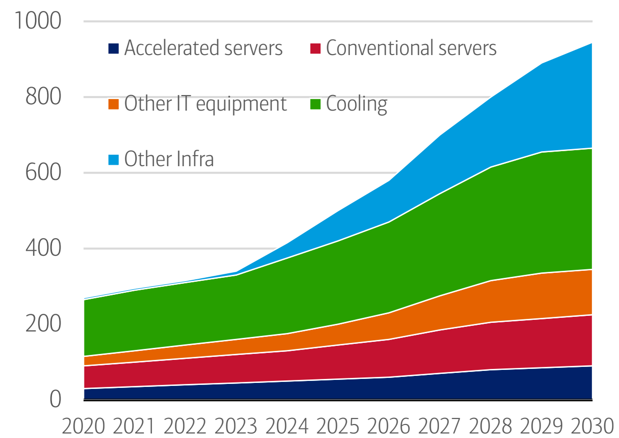
Exhibit 6: Cooling and servers account for 80% of power consumption
Source of data center electricity consumption, 2020-2030, IEA base case



Source: International Energy Agency (IEA), BofA Global Research estimates
Note: Other infra = other infrastructure.

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The bottleneck has shifted to firm, deliverable power

Power systems have historically been evaluated around energy adequacy or whether there is enough megawatt-hour (MWh) for the year. Data centers increasingly force the system to be evaluated around deliverability: location, firmness and controllability. That is why assets tied to access, not just generation, are being re-rated.

Once deliverability becomes a bottleneck, data centers stop being a simple demand driver and become a system-design problem: how much generation, wires, redundancy, storage and onsite infrastructure needs to be built; where does it sit; and who finances it. The DOE has already pointed to onsite generation, storage, innovative rate structures, transmission expansion and re-use of retired coal infrastructure as part of the response to data-center growth.

According to BofA Global Research, nuclear fits the data center load profile unusually well: very high-capacity factors, dense power close to site, long-duration contracts and low-carbon firmness. Data centers may therefore become the first credible commercial market for small modular reactors (SMRs), not because they are the cheapest source of power, but because they offer certainty of access in the most constrained markets, according to BofA Global Research (read publication [Q&A quick guide: Small modular reactors] for more).

Metals caught in the data center construction crosshairs

As AI reshapes data center architecture, metals are shifting from incidental construction inputs to strategic enablers of deployment. Rising compute density, advanced cooling architectures, and tighter uptime and redundancy standards mean metals demand is now driven by engineering constraints and system reliability, not floor space or headline capex.

Where metals exposure really sits

Although servers dominate the narrative, the bulk of metals in data centers sits outside the server hall, embedded in power delivery, redundancy and cooling infrastructure (Exhibit 9). Each incremental megawatt of capacity incorporates roughly 60-75 tons of metals, with power systems and cooling together accounting for 55-75% of total metals intensity.

Exhibit 7: Data centers would not work without metals
Mined raw materials used in data centers

Metals	Application
Structural and Rack Materials	
Steel	Used for server racks, frames, and building structures because of its strength and durability.
Aluminium	Common in lightweight racks and enclosures; also used for heat sinks due to its good thermal conductivity.
Electrical Conductors	
Copper	The primary metal for power cables, busbars, and networking cables because of its excellent electrical conductivity.
Aluminium	Sometimes used in power distribution for cost savings, though less conductive than copper.
Cooling and Heat Management	
Aluminium	Widely used in heat sinks and cooling components.
Copper	Used in high-performance heat exchangers and liquid cooling systems because of superior thermal conductivity.
Electronic Components	
Gold	Found in connectors and circuit boards for corrosion resistance and reliable conductivity.
Silver	Used in high-end contacts and solder for its excellent conductivity.
Tin	Common in solder for printed circuit board (PCB) assembly.
Nickel	Often used as a plating material for connectors.
Batteries and Backup Systems	
Lead	In lead-acid batteries for uninterruptable power system (UPS) systems.
Lithium	In lithium-ion batteries for modern backup solutions.
Specialty Metals	
Rare Earth Elements	Used in magnets for hard drives and cooling fans.

Source: BofA Global Research

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Copper: Small share, outsized marginal impact

Metals typically account for less than 5% of total data center capex, making demand highly price inelastic. As data center construction accelerates, copper demand is being pulled forward into an already tightening market, with AI projects acting as volatility amplifiers rather than marginal buyers.

By 2030, AI data centers could account for c.2% of global copper demand, up from <1% today, and add 2-3% to the North American copper demand compounded annual growth rate (CAGR) through 2035, driven mainly by grids and power equipment, according to IEA. In China, direct AI demand may reach 5-6%, while indirect grid-related demand could account for roughly half of copper demand growth.

Beyond copper: Aluminum, tin and specialty metals

But it's not just copper: aluminum demand from data centers is projected to more than double, from c.330 kiloton (kt) in 2025 to c.695 kt by 2030, according to BofA Global Research. Even small increases in demand can drive outsized price volatility when end demand is structurally inelastic and tied to fixed infrastructure build-out.

Beyond basic construction materials, data centers rely on a small number of specialized metals that are hard to source and often concentrated in a few countries. Supplies of key inputs like gallium, germanium and rare earths are sensitive to geopolitics and trade policy, which can quickly drive up prices and cause delays. Even though these materials are used in tiny amounts, they are essential to high-performance computing components (i.e. chips, fiber optics, power systems, cooling equipment) making the data-center supply chain unusually exposed to disruptions.

Geopolitics turns metals into strategic assets

Geopolitics has reclassified data-center metals into strategic infrastructure assets, according to BofA Global Research. In 2025, the US expanded Section 232 tariffs, lifting steel and aluminum duties to 50% and signaling potential further action on copper, increasing uncertainty around delivered costs and procurement. At the same time, China's dominance of refining and midstream capacity across copper, aluminum, electrical steel and silver has sharpened concerns over security of access.

Which metals bind first varies by region: transformers in the US, grid reinforcement in Europe, midstream concentration in China (Exhibit 10). Beneath the near-term volatility though, the structural story remains intact: global grid expansion, electrification and renewables-led power growth continue to push key markets like copper and aluminum into medium-term deficit.

Exhibit 8: No region can yet achieve full security of supply

Security of supply of energy transition metals and minerals across the transition metals outlook regions

	China		Europe		US		Japan and South Korea		Southeast Asia		Rest of world	
	2025	2035	2025	2035	2025	2035	2025	2035	2025	2035	2025	2035
Aluminum	100%	100%	67%	50%	70%	57%	78%	57%	100%	100%	100%	100%
Cobalt	9%	15%	9%	28%	3%	11%	1%	4%	100%	100%	100%	100%
Copper	17%	18%	100%	98%	86%	62%	36%	30%	100%	56%	100%	100%
Graphite	100%	100%	19%	19%	10%	19%	98%	64%	96%	33%	28%	16%
Lithium	49%	92%	4%	22%	9%	22%	2%	4%	0%	0%	100%	100%
Manganese	7%	7%	0%	0%	0%	2%	0%	1%	0%	0%	100%	100%
Nickel	10%	12%	55%	64%	39%	42%	21%	23%	100%	100%	100%	100%
PGMs	20%	30%	2%	0%	31%	40%	15%	20%	27%	26%	100%	100%
Steel	100%	82%	80%	72%	92%	76%	100%	100%	70%	100%	100%	100%

Key: Regional supply meets total demand | Regional supply meets most demand | Regional supply meets some demand | Regional supply meets little demand | Regional supply meets less than 40% of demand

Source: BloombergNEF. BofA Global Research

Note: The table assumes that all supply is consumed domestically. Demand is total demand (energy transition and other) under BNEF's Economic Transition Scenario. Supply is both primary and secondary supply. Cobalt, copper, lithium, manganese, nickel and platinum group metals (PGMs) are mined supply. Aluminum, graphite and steel are refined supply.

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Disclosures